

Last name _____

First name _____

LARSON—MATH 310—HOMEWORK WORKSHEET 07

Test 1 Review

1. Write up a **neat** assignment on a **new sheet** of paper. (Do not cram your answers between the lines).
2. **Number** your problems so that it is easy to see what work matches the assigned problems.
3. Be verbose. Remember that you do not understand a concept if you do not know an **examples**.

Definitions and Notation. Give a definition of each concept **and** an example.

1. What is $\mathbb{R}^n$?
2. What is the *norm* of a vector $\vec{v}$?
3. What is a *linear combination* of vectors?
4. What is the definition of the *linear dependence* of vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_k$?
5. What is the *inner product* of vectors $\vec{v}$ and $\vec{u}$?
6. What is the *distance* between vectors $\vec{v}$ and $\vec{u}$?
7. What is the angle $\angle(\vec{a}, \vec{b})$ between vectors $\vec{a}$ and $\vec{b}$?
8. What is a *basis*?
9. What does it mean for a set of vectors to be *orthogonal*?
10. What does it mean for a set of vectors to be *orthonormal*?
11. What is a *diagonal* matrix?

Theorems

12. What is the Cauchy-Schwartz Inequality? Give an example.

Algorithms

13. Explain the Gram-Schmidt algorithm.

Problems

14. Let $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$. Find $\|\vec{v}\|$.

15. Find $\vec{1}^T \vec{v}$ and $\vec{v}^T \vec{1}$ for $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$.
16. Check that vectors $\vec{a}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\vec{a}_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are linearly dependent.
17. Check that $\vec{e}_1, \vec{e}_2, \vec{e}_3, \vec{e}_4$ are linearly independent (in $\mathbb{R}^4$).
18. What is the *orthonormal expansion* of $\vec{x} = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$ with respect to the orthonormal list $\vec{e}_1, \vec{e}_2, \vec{e}_3$?
19. Find the *inner product* of $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$.
20. Explain why the function $f(x, y) = 2x + 3y$ is linear.
21. How can we write the linear function $f(x, y) = 2x + 3y$ as an inner product?
22. Show that $\vec{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $\vec{a}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ are linearly independent using the **definition** of linear independence.
23. Show that $\vec{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $\vec{a}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ are linearly independent using the **Gram-Schmidt algorithm**.
24. Let
- $$A = \begin{bmatrix} 0 & 1 & 2 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$
- (a) Why can you instantly say that the columns are linearly dependent?
- (b) What is the largest possible number of linearly independent columns? (Cite the appropriate result from class. Explain.).
- Let M be the 3×2 matrix: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$
25. Find M^T .
26. Why isn't MM well-defined?
27. Find MM^T .
- Let $\vec{x} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$.
28. Use the *column interpretation* of matrix-vector multiplication to find $M\vec{x}$.
29. Find a 3×3 *upper triangular* matrix U with all non-zero entries equal to 2.